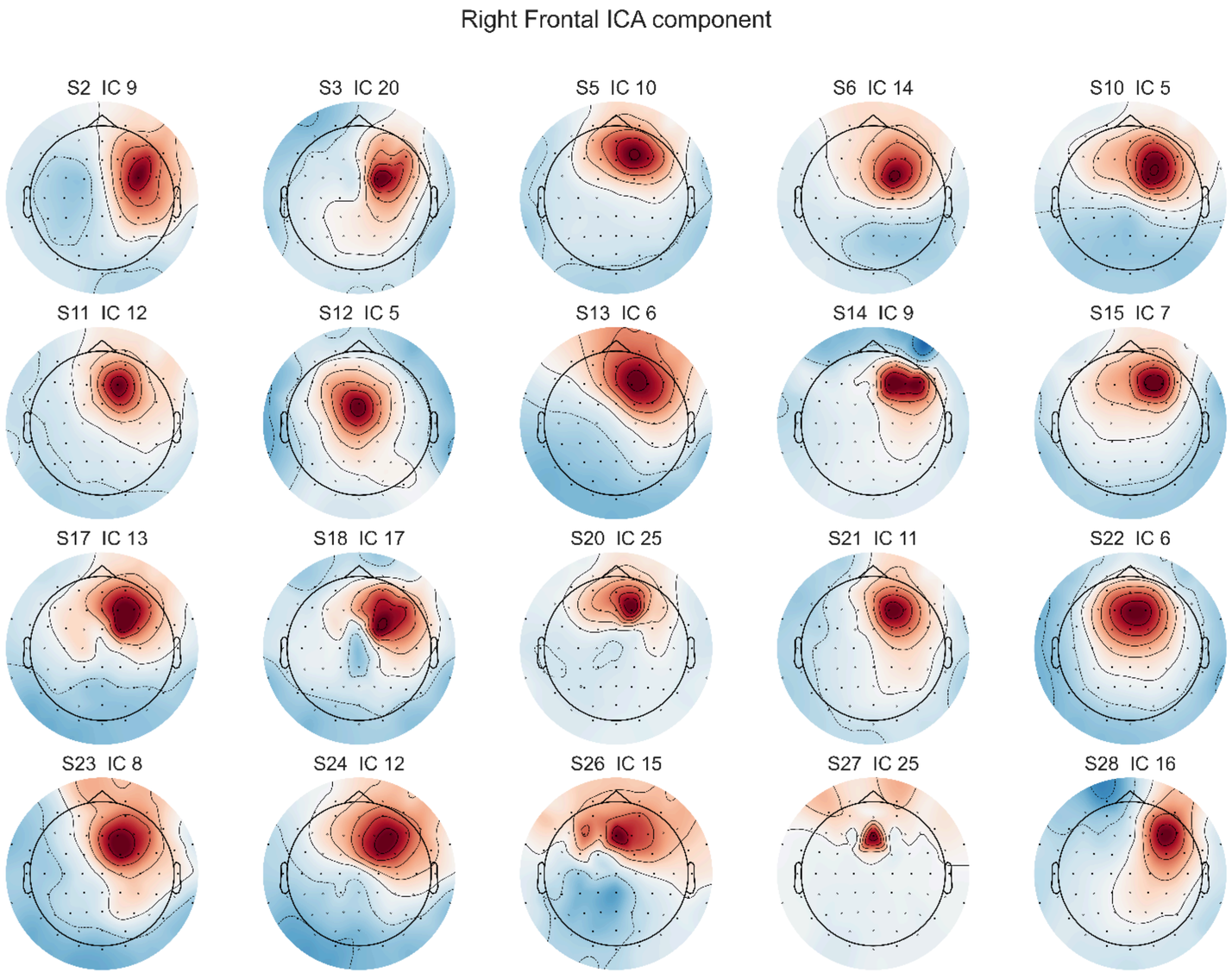
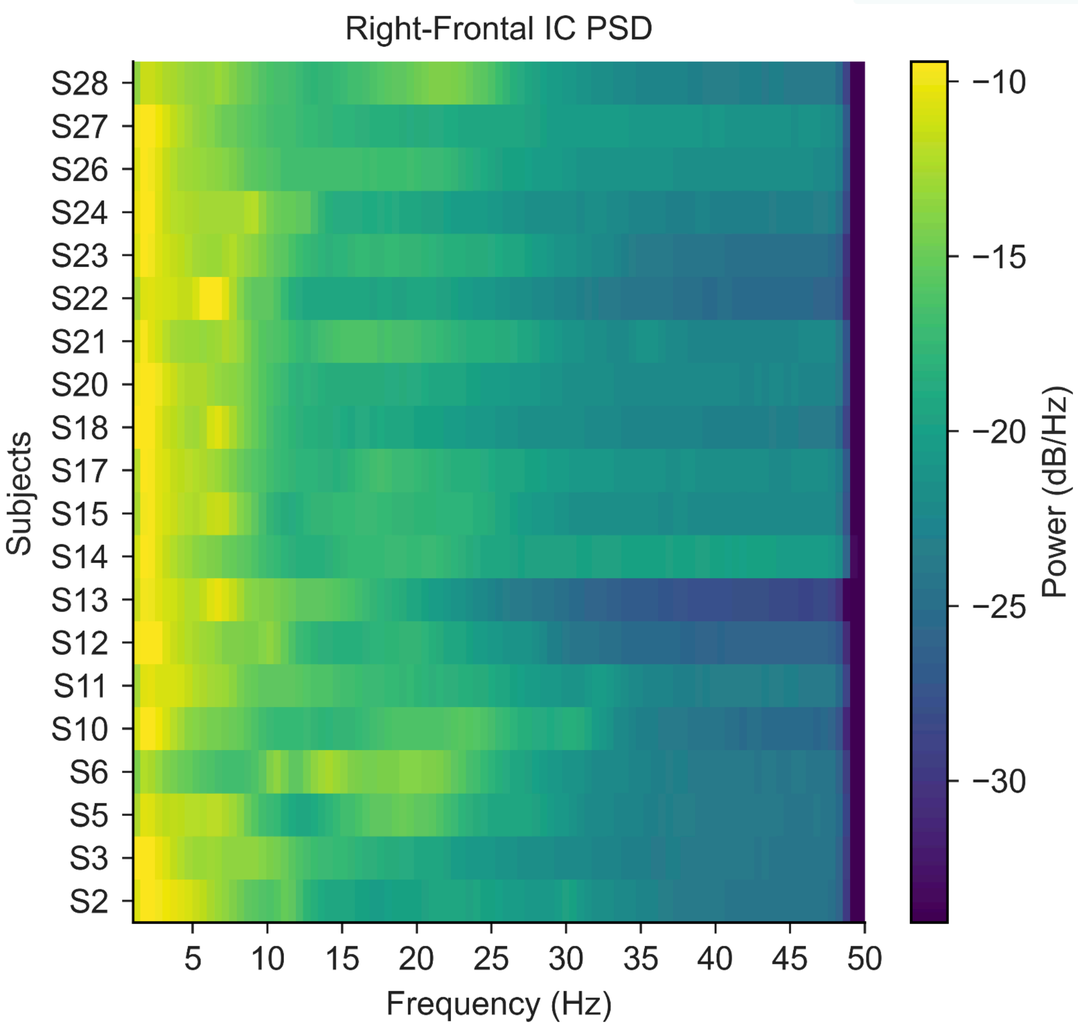
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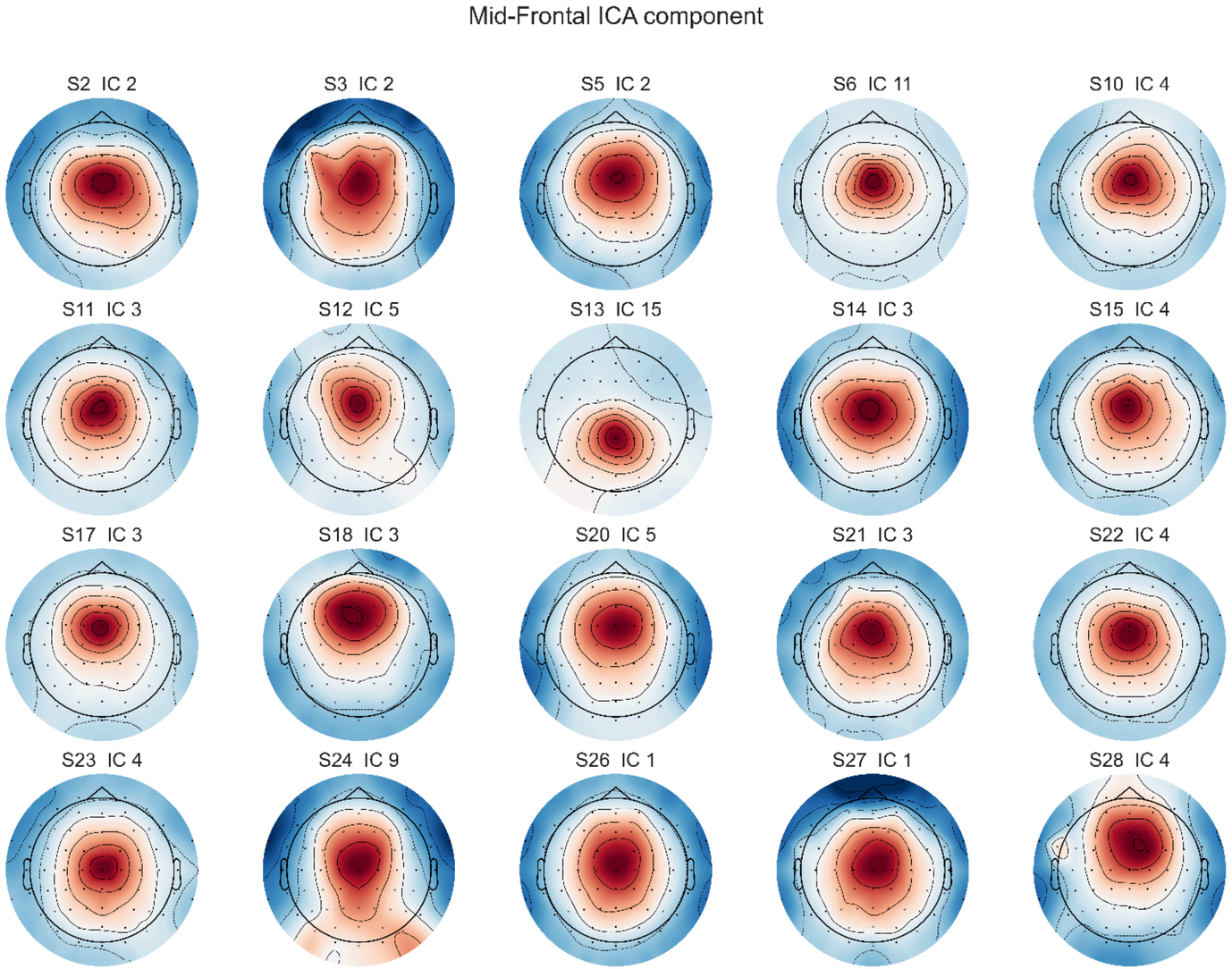
Shoto Yamada

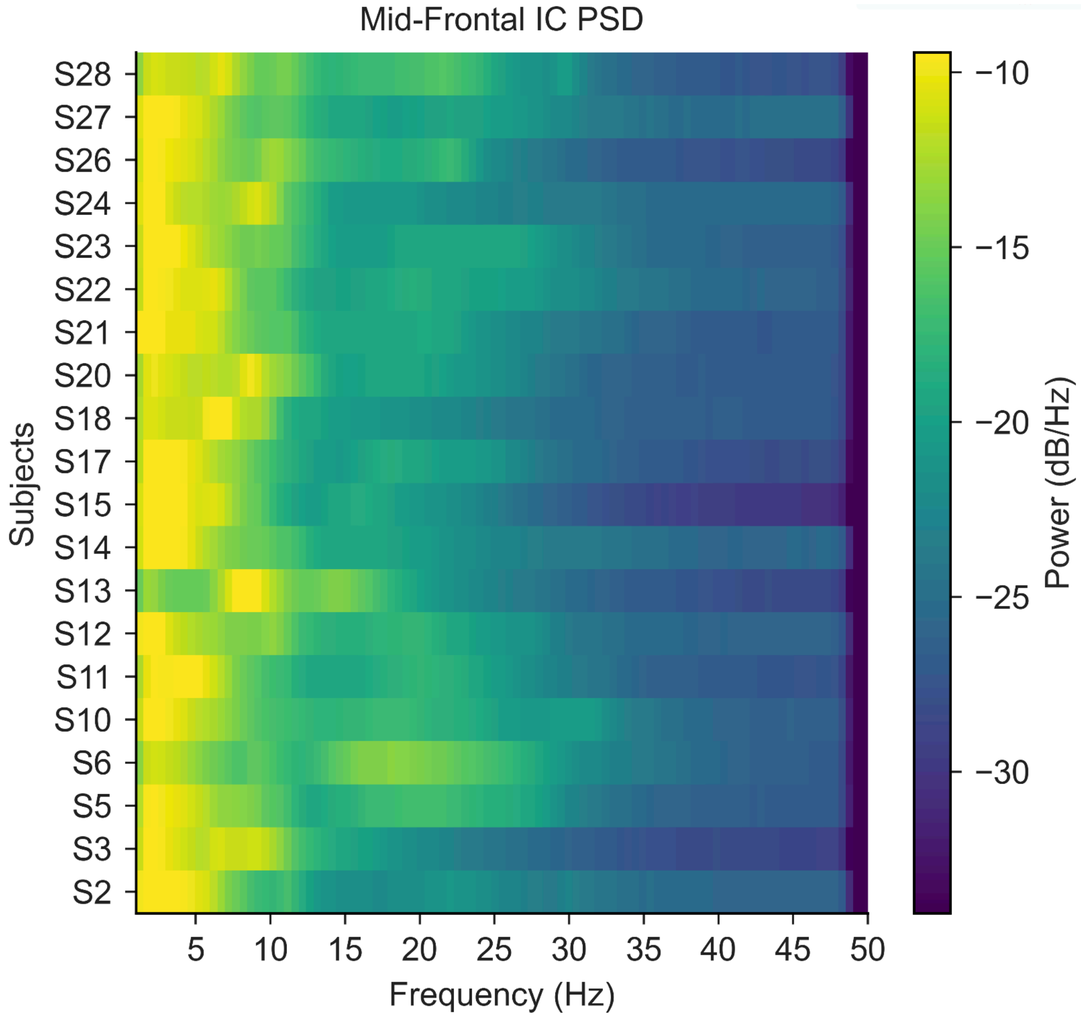
Methods

- Right-frontal and mid-frontal ICA components were identified for each subject.
- The component time series were downsampled from 500 Hz to 100 Hz using MNE.
- TDE-DP-GMM was then applied to the data.
- For model fitting, the right-frontal and mid-frontal IC time series were concatenated into one long signal.
- For subject 12, the shared IC component was included only once, because the same component was used for both the right-frontal and mid-frontal IC.
- Candidate frequencies of 5, 10, 20, and 30 Hz were considered, corresponding to 17 embeddings (154 ms window length).
- The model was fit with an upper bound of 16 states.
- All signals were z-scored before model fitting.
- After state estimation, post-hoc k-NN imputation was applied using 7 samples (70 ms), corresponding approximately to 2 cycles at 30 Hz.
- The estimated state sequences were then epoched into trials for each subject and IC.


Right Frontal ICA component
S2 IC 9
S3 IC 20
S5 IC 10
S6 IC 14
S10 IC 5
S11 IC 12
S12 IC 5
S13 IC 6
S14 IC 9
S15 IC 7
S17 IC 13
S18 IC 17
S20 IC 25
S21 IC 11
S22 IC 6
S23 IC 8
S24 IC 12
S26 IC 15
S27 IC 25
S28 IC 16


Right-Frontal IC PSD
Subjects
S28
S27
S26
S24
S23
S22
S21
S20
S18
S17
S15
S14
S13
S12
S11
S10
S6
S5
S3
S2
5
10
15
20
25
30
35
40
45
50
Frequency (Hz)
−10
−15
−20
−25
−30
Power (dB/Hz)


Mid-Frontal ICA component
S2 IC 2
S3 IC 2
S5 IC 2
S6 IC 11
S10 IC 4
S11 IC 3
S12 IC 5
S13 IC 15
S14 IC 3
S15 IC 4
S17 IC 3
S18 IC 3
S20 IC 5
S21 IC 3
S22 IC 4
S23 IC 4
S24 IC 9
S26 IC 1
S27 IC 1
S28 IC 4


Mid-Frontal IC PSD
Subjects
S28
S27
S26
S24
S23
S22
S21
S20
S18
S17
S15
S14
S13
S12
S11
S10
S6
S5
S3
S2
5
10
15
20
25
30
35
40
45
50
Frequency (Hz)
−10
−15
−20
−25
−30
Power (dB/Hz)

Experiment overview

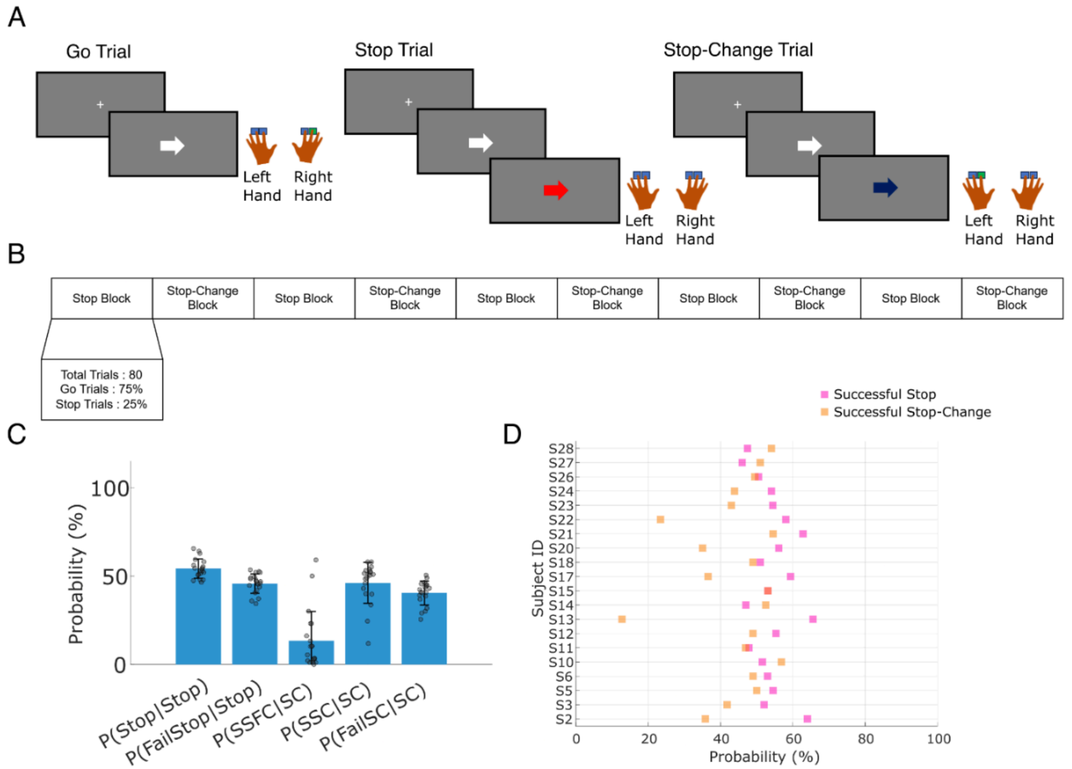


Figure 1: Overview of the stop-change task. (A) Schematic illustration of the three types of trial. Each trial begins with a fixation followed by a Go cue (white arrow). In go trials, participants responded with the right hand. In stop trials, the go cue was followed by a stop signal (red arrow) instructing the participant to inhibit the planned response. In stop-change trials, the stop signal was replaced by a change cue (blue arrow), requiring the participants to cancel the prepared right hand response and press with their left hand. (B) Experimental block structure. The experiment consisted of alternating stop blocks and stop-change blocks, each containing 80 trials with a ratio of 75% of go trials and 25% stop or stop change trials. (C) Group-level behavioural performance across conditions. Bar plots represent the mean probability of successful stopping, failed stopping, successful stop and failed change, successful change and failed change. (D) Individual Participant performance for successful stop and successful stop - change.

Behavioral results

- No significant difference between SSRT and CSRT (?)
  - SSRT: 205.85 ms
  - CSRT: 206.77 ms

- My version:
  - SSRT: 210.12 ms
  - CSRT: 184.48 ms

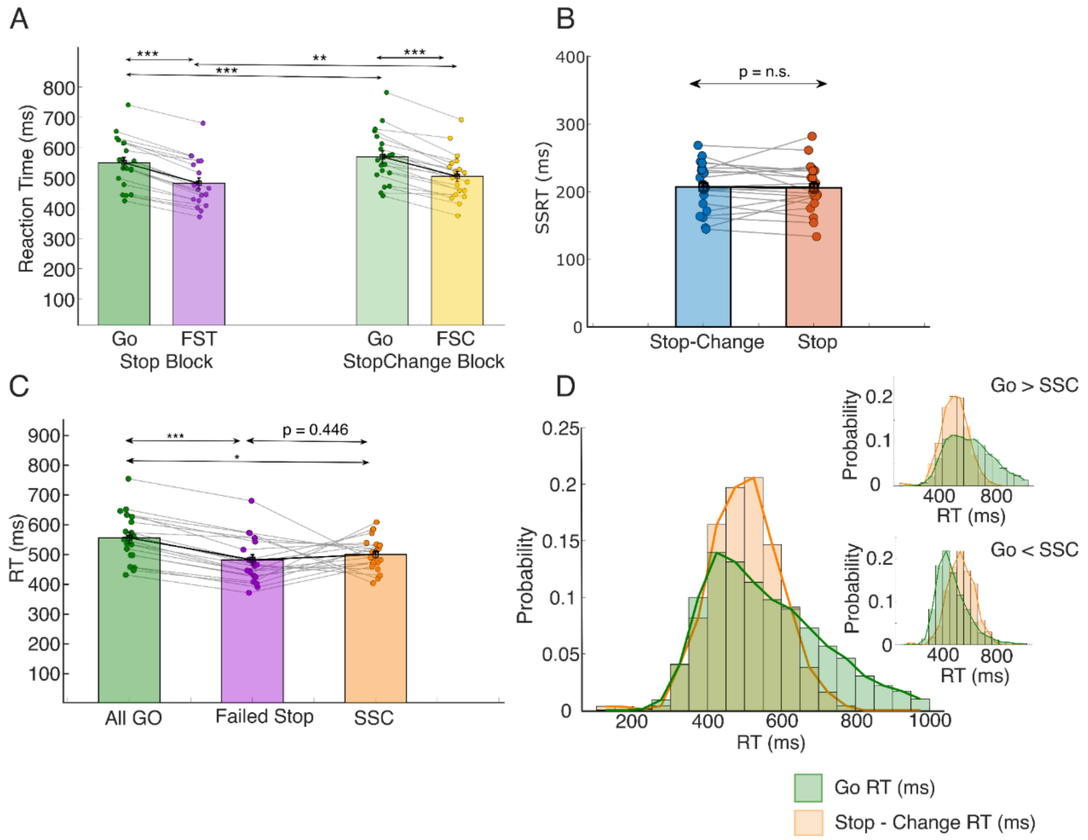


Figure 2: Behavioural Performance
(A) Mean Reaction time for Go trials in the stop block and fails stop trials (FST) as well as go trials from the stop-change block and failed stop change trials (FSC). (B) Stop-signal Reaction Time(SSRT) estimates of stop and stop-change conditions. No statistically significant difference (p = 0.862) (C) Comparison of the reaction times across all go trials, failed stop trials and successful stop-change trials. (D) Distribution of the reaction times for the Go trials and stop-change trials.

Behavioral results

Connection to the previous results

Overall, elongated CSRT is observed, but results are mixed

Table 1
Overview on studies using the stop-signal paradigm and the change paradigm in the same study.

Author	Design	n	Paradigm[a]	SSRT (M, SD)	CSRT (M, SD)	ES[b] (95%-CI)	StopRes. (M, SD)	ChangeRes. (M, SD)	ES (95%-CI)	go-RT (M, SD)
Logan and Burkell (1986)	Behavioral (between)[c]		Stop, change							
	Controls (exp 1)	12		204	305					
	Controls (exp 2)	6		206	250					
Drueke et al. (2010)	Pharmaceutical[d]		Stop-change							
	Controls (placebo1)	13		216 (48)	209 (57)	.13 (−.64–.90)				490 (138)
	Controls (placebo2)	15		222 (50)	226 (56)	.07 (−.79–.64)				416 (53)
Lijffijt et al. (2006)*	Pharm./patient (ADHD, child)	15/11	Stop, change (foot)							
	Placebo			350 (161)	359 (189)	.05 (−.73–.83)				S: 621 (90)/C: 667 (140)
	MPH (0.5 mg/kg)			251 (91)	296 (95)	.47 (−.32–1.26)				S: 580 (88)/C: 616 (102)
	MPH (1.0 mg/kg)			211 (93)	262 (75)	.57 (−.22–1.37)				S: 567 (89)/C: 623 (106)
Bekker et al. (2005)	Patient (ADHD, adult)	24	Stop, change (foot)	237 (87)	308 (88)	.80 (.21–1.39)				S: 468 (88)/C: 505 (105)
	Controls	24		185 (39)	246 (68)	1.08 (.48–1.69)				S: 463 (69)/C: 488 (91)
McClure et al. (2005)	Patient (BPD, child)	38/37	Stop, change	251 (71)	299 (84)	.61 (.15–1.07)				
	controls	22		217 (63)	253 (56)	.59 (.01–1.2)				
De Jong et al. (1995)	EEG controls	8	Stop, change (foot)	174	220		400	411		S: 443/C: 455
Boecker et al. (2007)	fNIRS, controls	15	Stop-change	234 (42)	245 (50)	.23 (−.95–.49)				546 (125)
Boecker et al. (2011)	fMRI, controls	15	Stop-change	214 (42)	238 (39)	.58 (−.15–1.31)	492 (139)	501 (135)	.06 (−.65–.78)	546 (171)
Krämer et al. (2011)	EEG, controls	15	Combined stop-change/Flanker	246 (31)	224 (27)	−.74 (−1.48–0)	364 (39)	361 (38)	−.08 (−.79–.64)	c: 401 (51)/ic: 424 (51)

Note. N: number; SSRT: stop signal reaction time; CSRT: change signal reaction time; M: mean; SD: standard deviation; ES: effect size; CI: confidence interval; go-RT: speed of responding on correct go-trials; StopRes.: speed on stop-trials for which the participants did not stop; ChangeRes.: speed on change-trials for which the participants did not change; exp: experiment; pharm.: pharmaceutical; ADHD: attention deficit/hyperactivity disorder; MPH: methylphenidate; S: stop-signal task; C: change-task; BPD: bipolar disorder; c: compatible trials; ic: incompatible trials.

[a] Stop, change: both tasks were presented separately; stop-change: stop-trials and change-trials were presented within the same paradigm; if not stated otherwise, the change response had to be performed manually.

[b] Following the recommendations of Dunlap et al. (1996) the effect size was calculated for independent variables instead of dependent variables as effect sizes for dependent variables often overestimate the actual size of effect. Effect sizes were calculated as the difference of the means divided by the pooled standard deviation: $d = (m_1 - m_2)/[(SD_1 + SD_2)/2]$ and were bias-corrected (hedges).

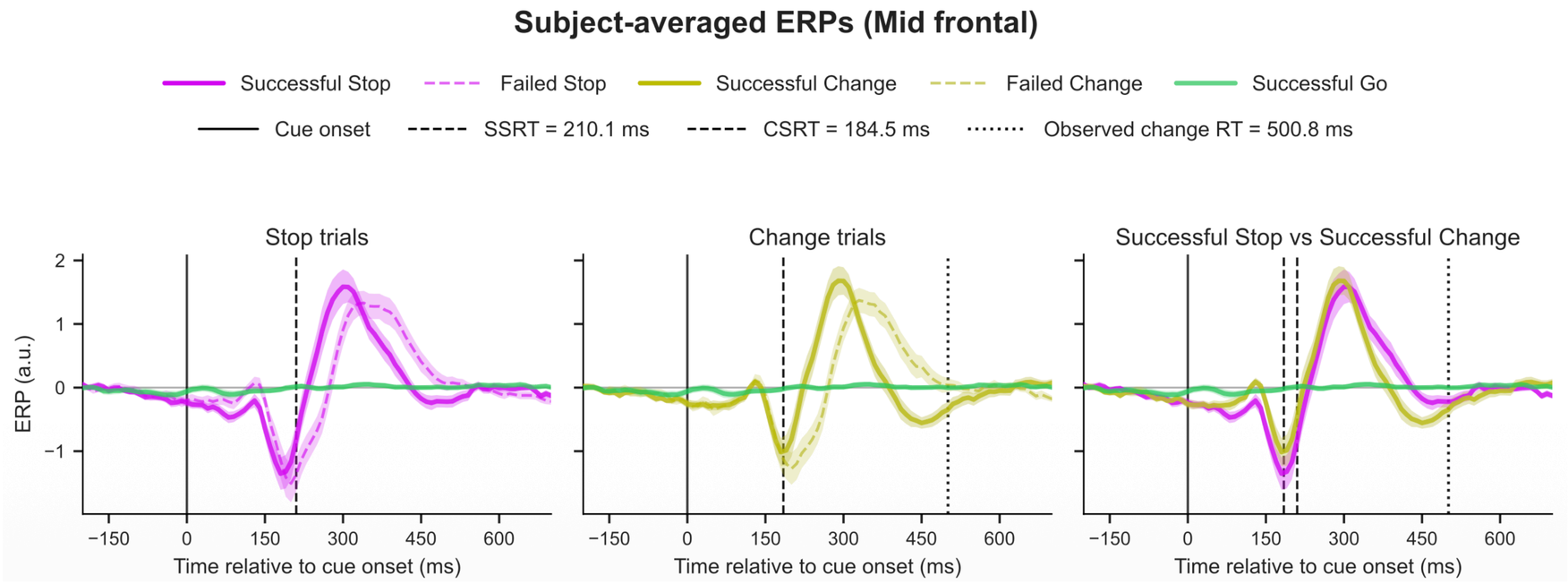
[c] The Logan and Burkell-study is the only study using a between-subject design.

[d] Placebo1: acute intake; placebo2: repeated intake.

Boecker, M., Gauggel, S., & Drueke, B. (2013). Stop or stop-change—Does it make any difference for the inhibition process? International Journal of Psychophysiology, Electrophysiological and Neuroimaging Studies of Cognitive Control: Evidence from Go/NoGo and Other Executive Function Tasks, 87(3), 234–243. https://doi.org/10.1016/j.ijpsycho.2012.09.009

EEG results: event-related potential

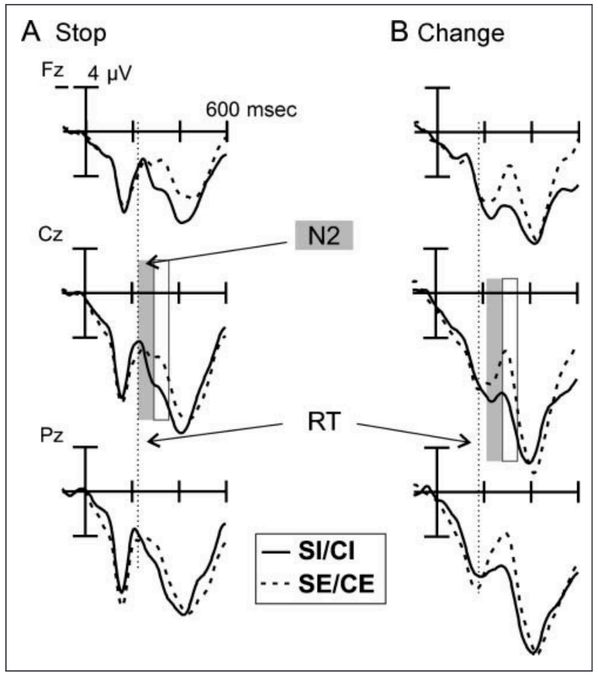
- N200 and P300 were observed in both stop and change
- For both successful stop and change, P300 onset was earlier and its amplitude was higher
- P300 onset and its amplitude seems the same
- N200 amplitude, however, looks higher in stop



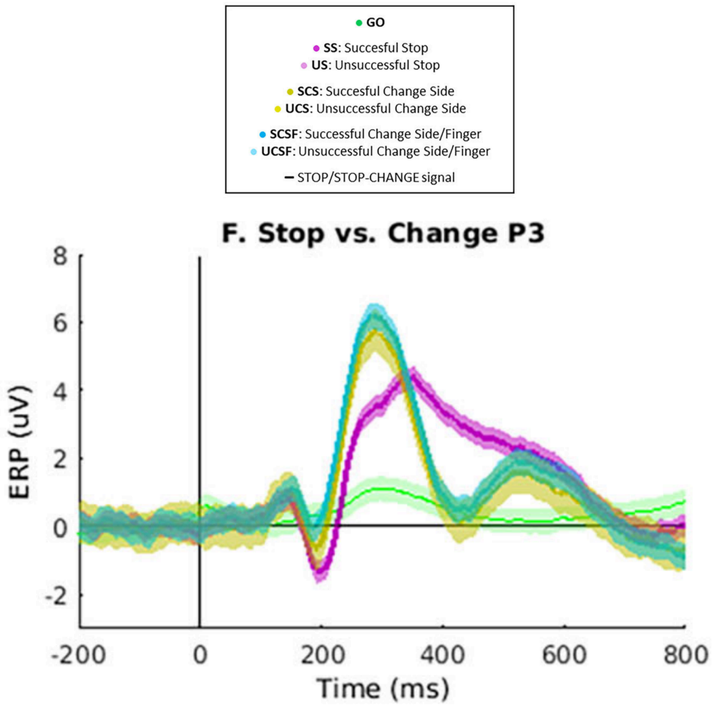
EEG results: event-related potential

Connection to the previous results

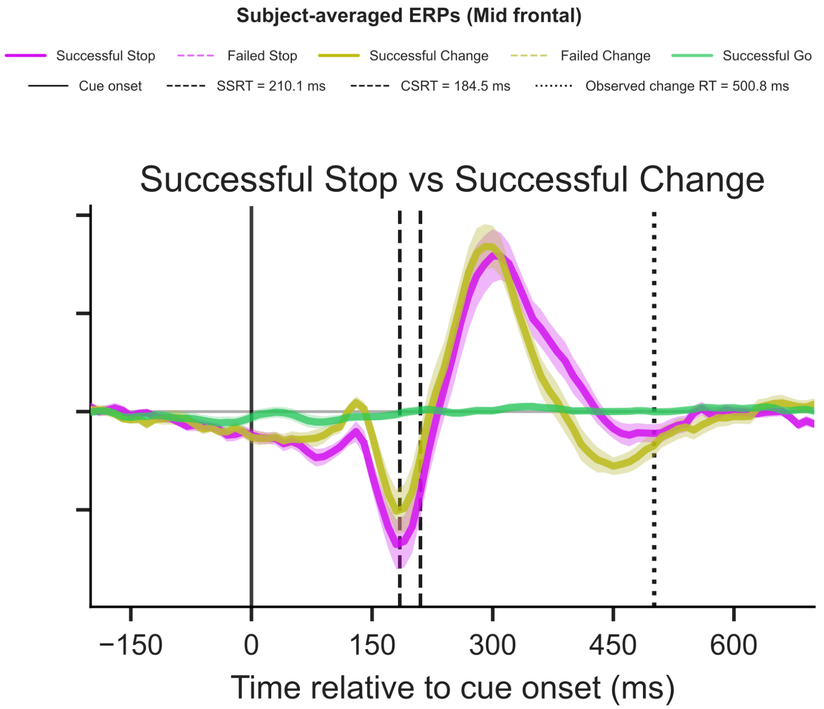
N200 was found only in stop (Krämer et al 2011)



N200 was not reported at all, but its amplitude was lower (i.e. larger deflection) in stop than change (Hervault and Wessel, 2025)



N200 looks lower in stop than change (Ours)

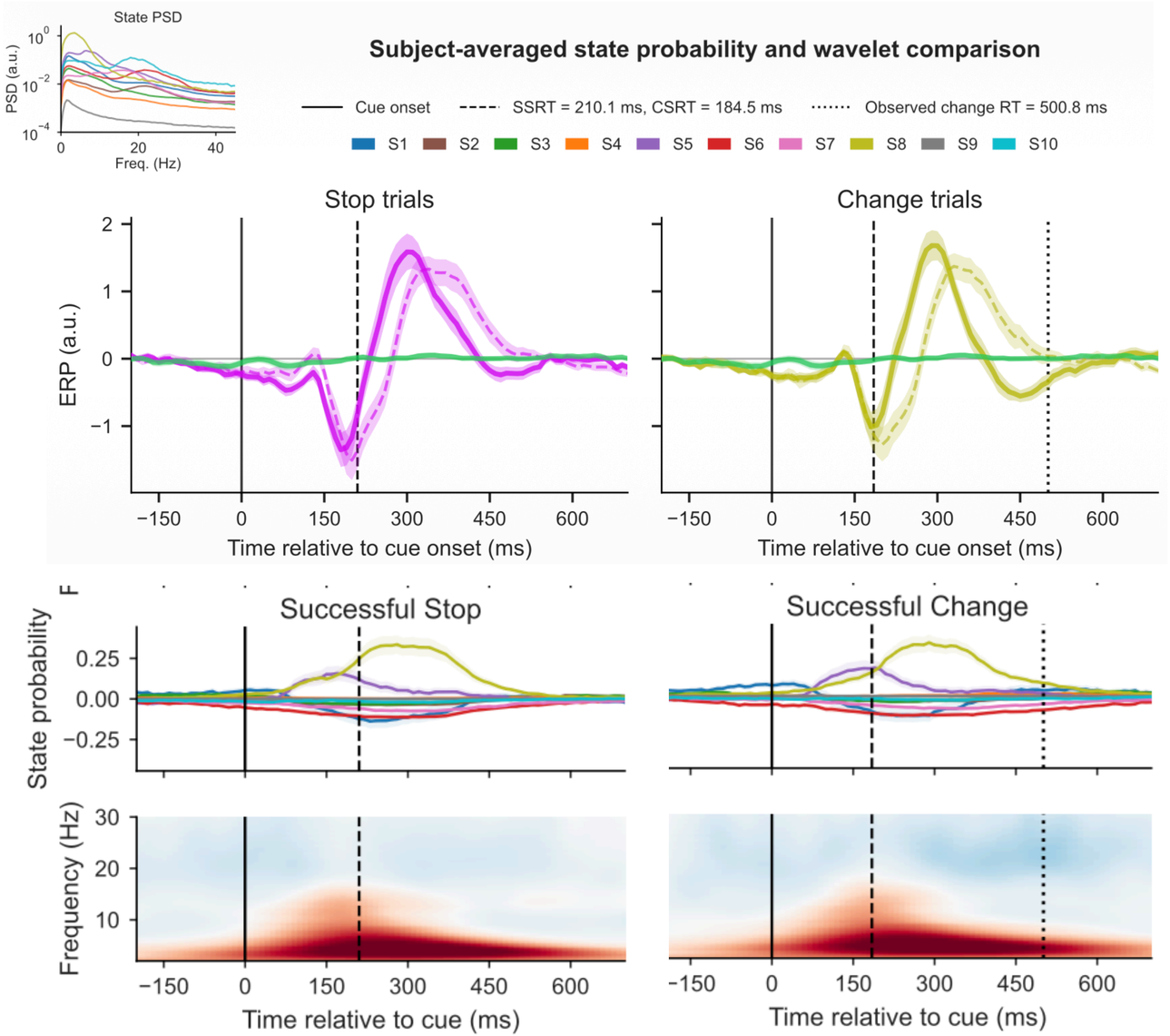


Krämer, U. M., Knight, R. T., & Münte, T. F. (2011). Electrophysiological Evidence for Different Inhibitory Mechanisms When Stopping or Changing a Planned Response. *Journal of Cognitive Neuroscience*, *23*(9), 2481–2493. https://doi.org/10.1162/jocn.2010.21573

Hervault, M., & Wessel, J. R. (2025). Common and Unique Neurophysiological Processes That Support the Stopping and Revising of Actions. The Journal of Neuroscience, 45(13), e1537242025. https://doi.org/10.1523/JNEUROSCI.1537-24.2025

EEG results: ERP and state-analysis

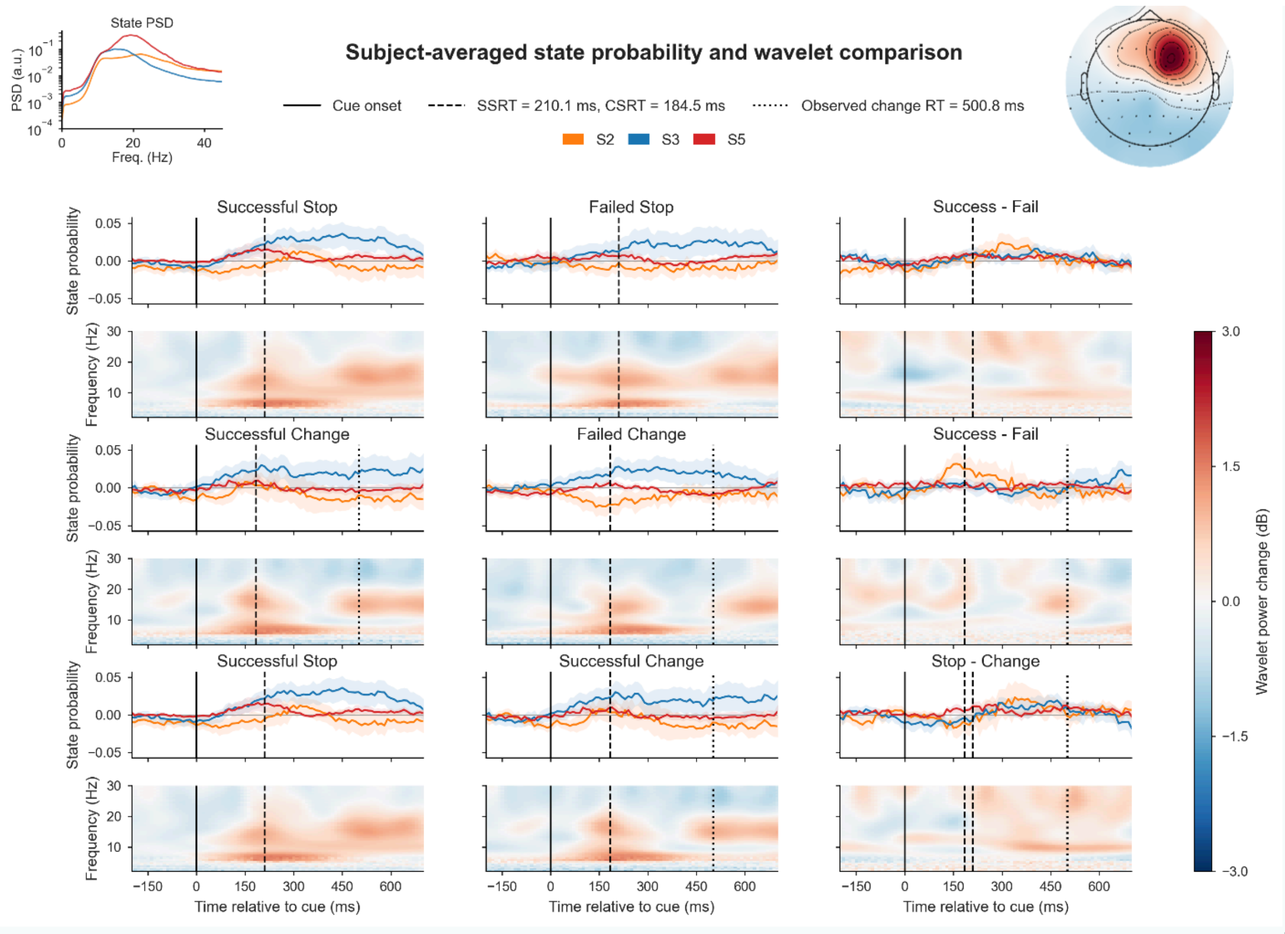
- ERP dominates the signal, such that identified states mostly reflect states
  - This justifies the use of high-pass filter for state-analysis
- Probability of ERP (?)
  - S5 (purple state) reflects N200
  - S8 (yellow state) reflects P300

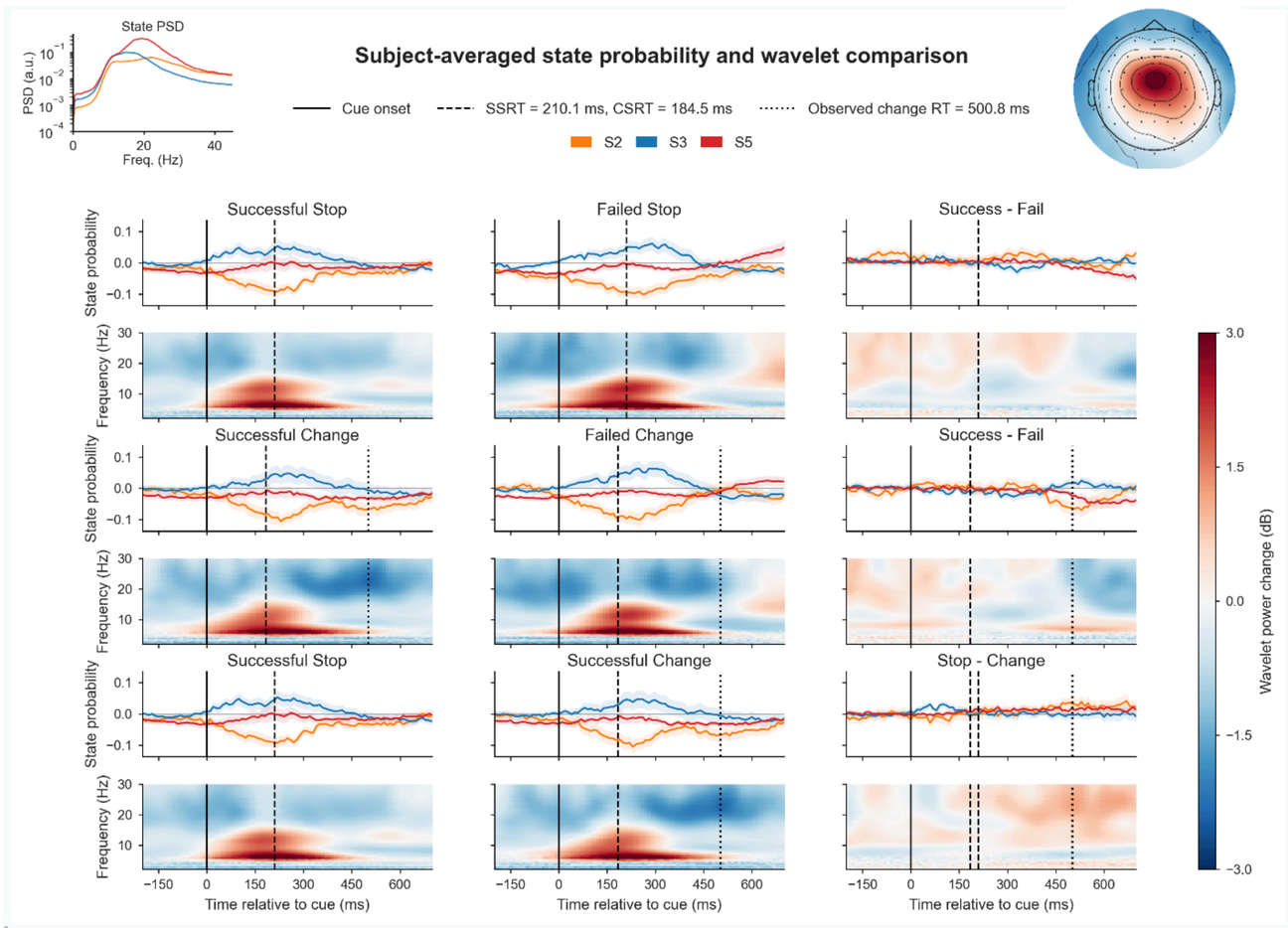


EEG results: wavelet and state-space analysis

High-pass filtered (>10Hz); beta states

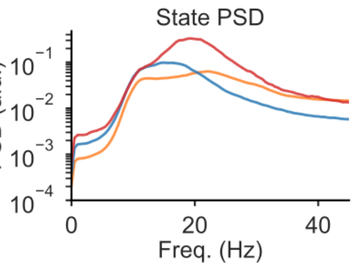
❖Stop and change showed similar state time course

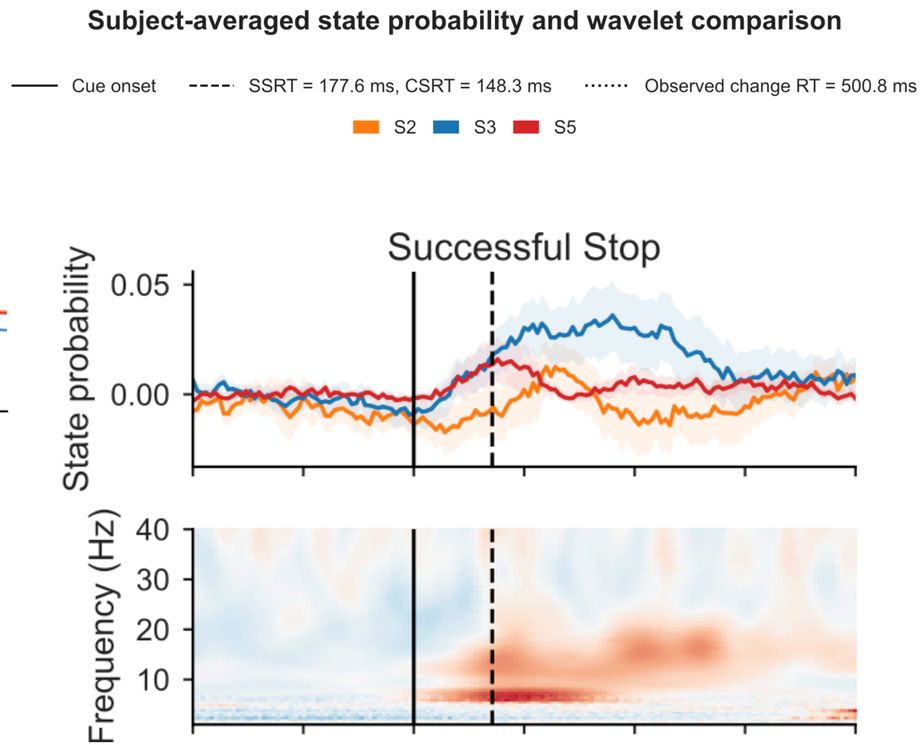


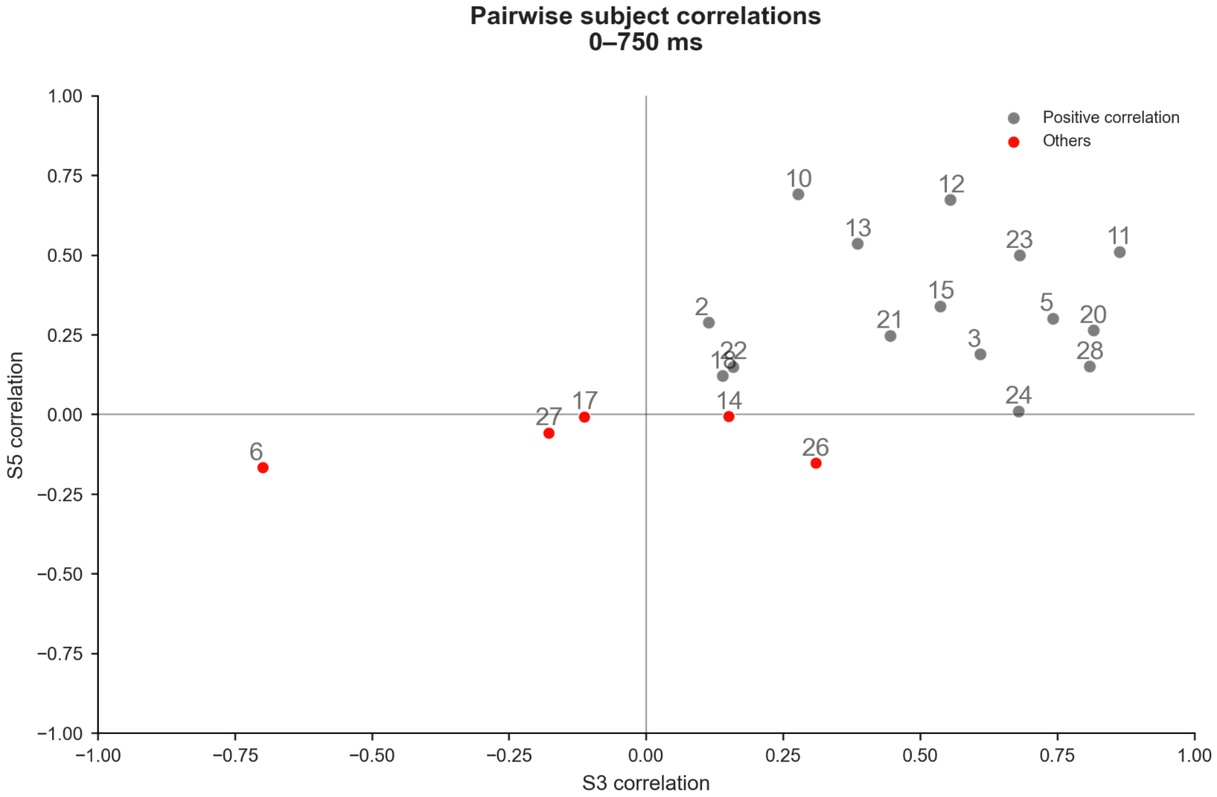


EEG results: grouping subjects

❖5 participants showed unique state time course from the other 15 participants

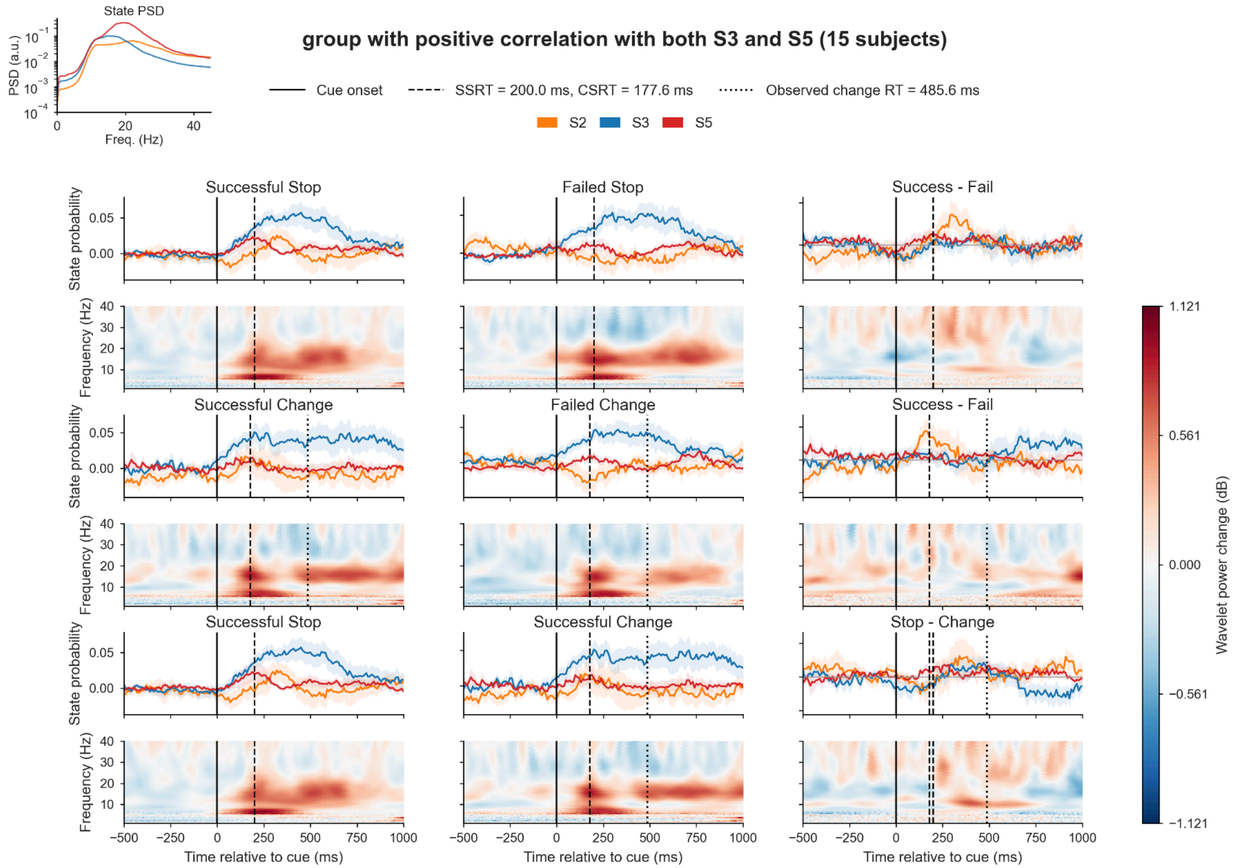


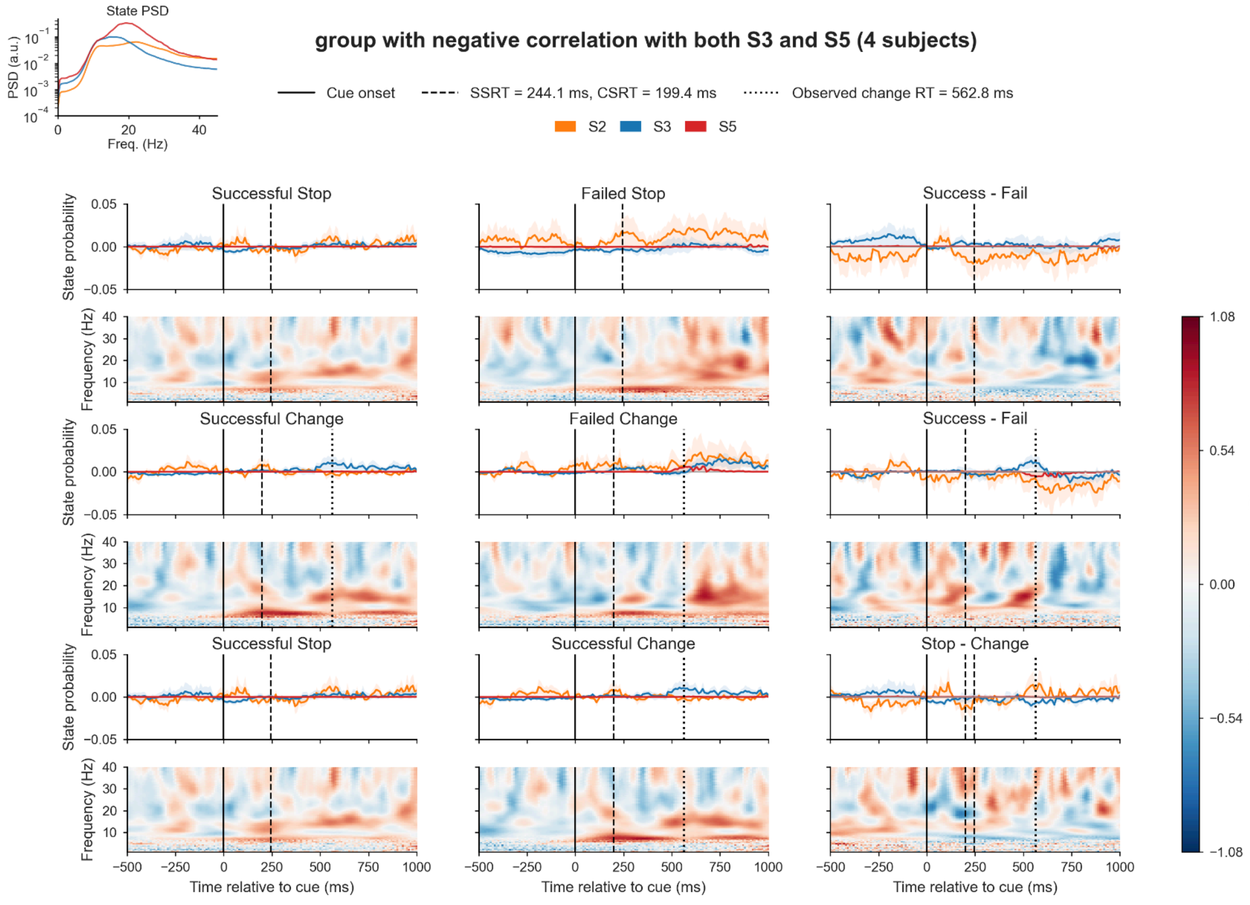




EEG results: grouping subjects

- ❖4 outlier participants showed little burst state activation
- ❖Time-frequency plot confirmed this
- ❖Notably, SSRT and CSRT are longer in this group
- ❖Some of them showed change RT > go RT



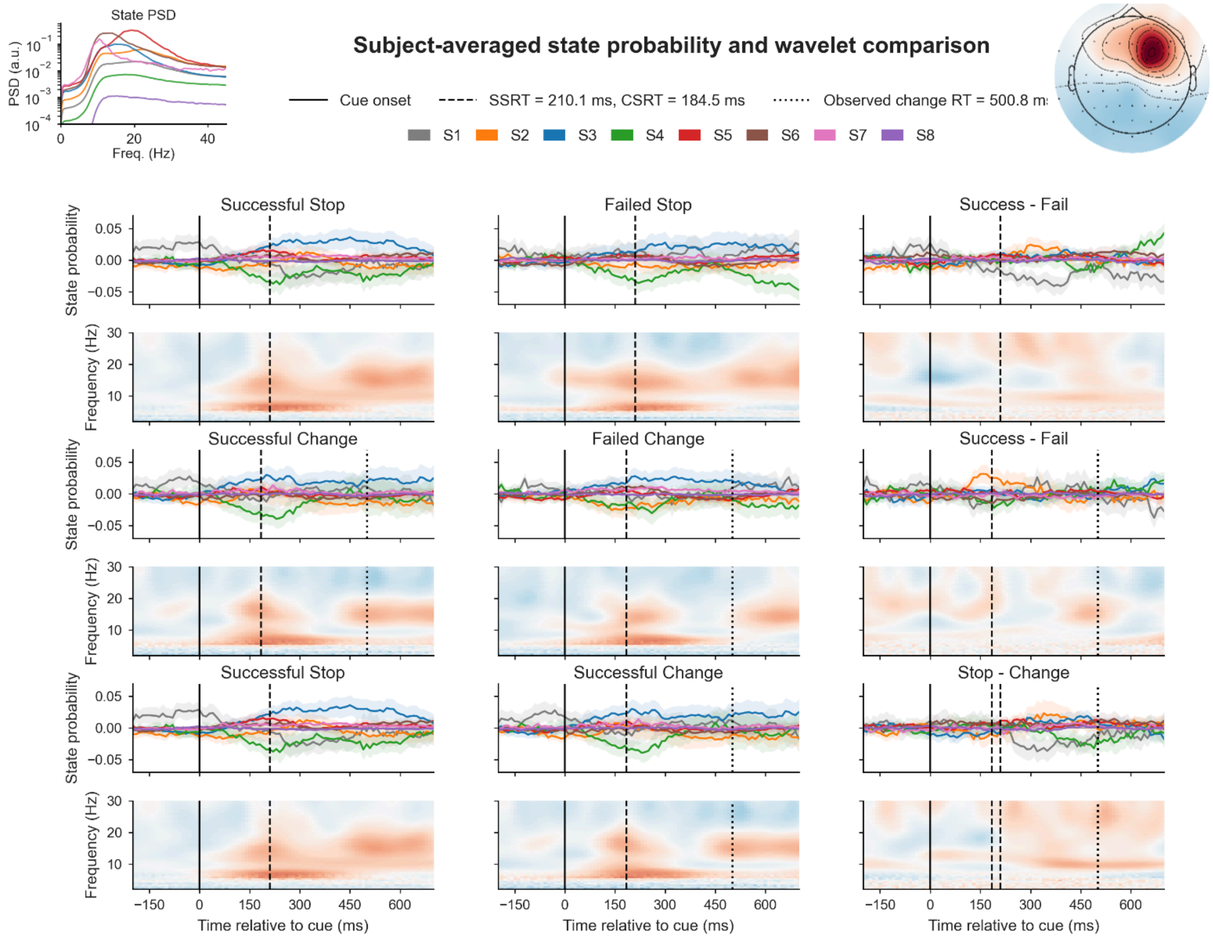


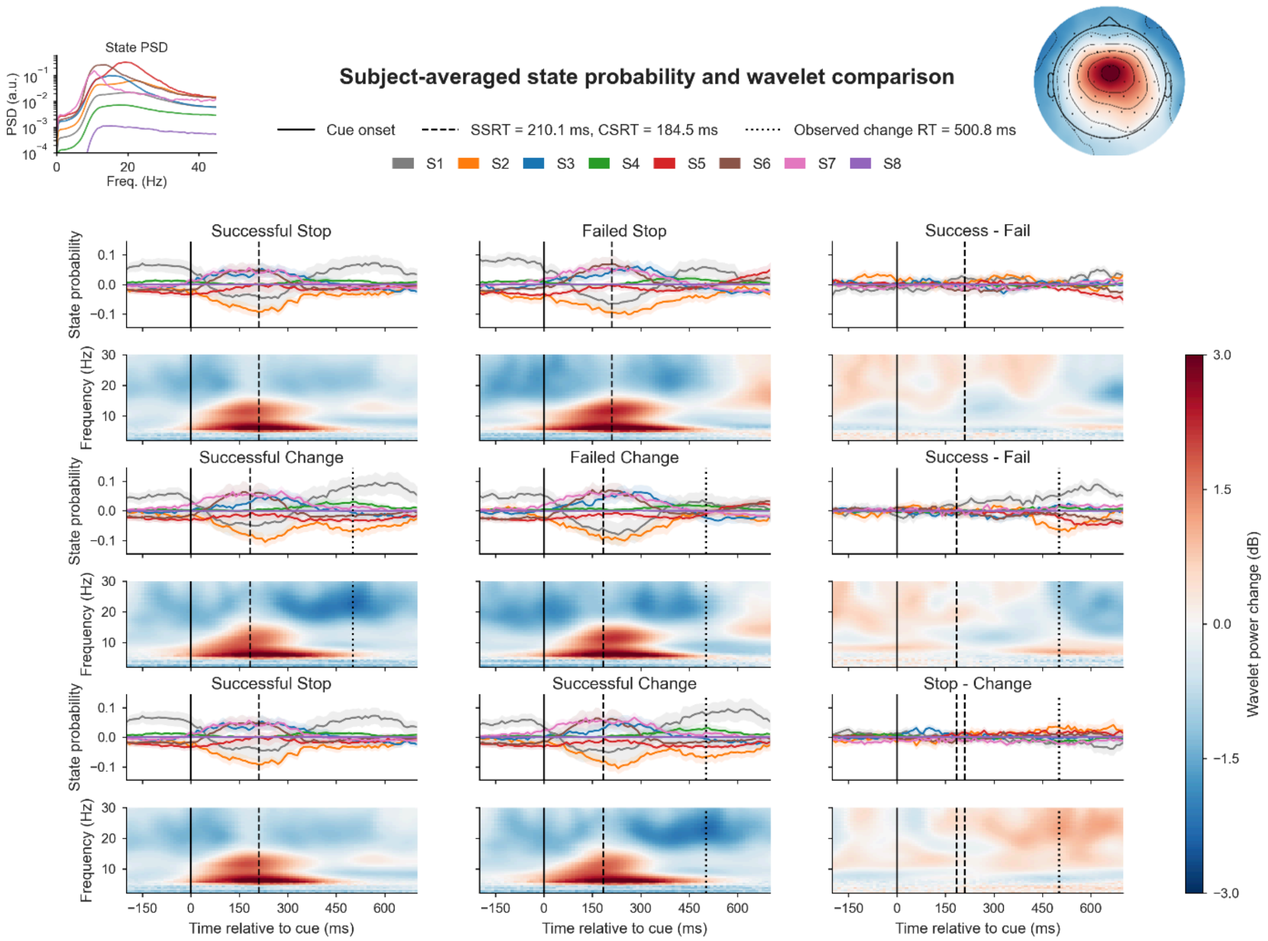
Supplements

EEG results: wavelet and state-space analysis

High-pass filtered (>10Hz); all states

❖Stop and change showed similar state time course





EEG results: wavelet and state-space analysis

Unfiltered signals ; all states

❖Stop and change showed similar state time course

